

CourtVision: Evaluating Vision-Guided Frame Selection for Basketball Video Question Answering

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Abstract—Video question answering requires a model to identify relevant visual evidence while also understanding events that unfold over time. Processing every frame of a video can be computationally expensive, making frame selection an important component of multimodal video understanding systems. This work presents CourtVision, a basketball video understanding system that compares three frame-selection strategies under a fixed eight-frame input budget: uniform temporal sampling, motion- and object-guided sampling using YOLO, and question-aware sampling that additionally incorporates CLIP image-text similarity. All three methods are evaluated using the same Qwen3-VL-8B-Instruct model on a fixed subset of 60 basketball questions from the Sports-QA dataset. Uniform sampling achieved 25.0% exact-match accuracy and 0.3961 mean Token F1. Motion+YOLO matched the exact-match accuracy but produced a slightly lower Token F1 of 0.3929, while adding CLIP reduced performance to 21.67% exact match and 0.3641 Token F1. The results suggest that visually salient or semantically relevant individual frames do not necessarily preserve the temporal evidence required for sports VideoQA.

Index Terms—Video question answering, basketball, frame selection, multimodal models, YOLO, CLIP, sports video analysis.

I. INTRODUCTION

Recent multimodal vision-language models can answer natural-language questions regarding images and videos. Yet video understanding presents an extra difficulty, since a video may have hundreds or thousands of frames, while a multimodal model can only practically deal with a limited amount of visual information at one time. A typical approach is to select a small number of frames from the entire video before passing them on to the model. The quality of the frames that are chosen can have a direct impact on whether the model gets enough evidence to answer the question.

Uniform temporal sampling is one of the most straightforward methods, since frames are taken at roughly equal intervals over the entire video. While this approach gives a wide temporal coverage, it does not take into account whether or not the individual frames contain important actions or objects. In contrast, more advanced methods can choose frames according to visual activity or in relation to a question. Recent studies have demonstrated an increasing interest in adaptive and query-aware frame selection for Video-LLMs [3], [4].

Basketball is a good situation in which to study this problem, since significant events are dynamic and usually depend on temporal relationships. A question like “What did the player do?” can be answered by referring to a single representative frame, whereas a question such as “Why did the player move towards the basket?” would need evidence from several moments both before and after the action. The Sports-QA benchmark was created specifically for VideoQA in the field of professional sports and contains descriptive, chronological, causal, and counterfactual questions [1].

The project looks into whether frame selection guided by computer vision gives stronger evidence for basketball VideoQA than a simple uniform baseline. Three methods are compared, with the downstream multimodal model and the number of frames remaining unchanged. The first method makes use of eight frames that are evenly spaced. The second method combines frame-to-frame motion with detections of players and the basketball based on YOLO. The third method builds on the second by incorporating CLIP similarity between each candidate frame and the question.

The primary research question is:

Does selecting frames based on visual and question-aware criteria give better results for basketball VideoQA than using uniform temporal sampling when the budget is fixed at eight frames?

Another aim of CourtVision is to show how its computer-vision features can be used to support basketball analytics. For this purpose, a separate prototype module carries out player tracking, trajectory analysis, player-spacing measurements, and occupancy visualization on the same basketball videos. The analytics are regarded as an application extension, not as part of the main VideoQA evaluation.

The main contributions of this project are:

1. A controlled comparison of three frame-selection strategies—those based on uniformity, those guided by motion or objects, and those that are question-aware—when applied to basketball VideoQA.
2. An analysis of the different ways in which these strategies behave when applied to descriptive and causal questions.
3. A prototype basketball analytics pipeline which makes use of player tracking, movement statistics, and visual occupancy analysis.

II. RELATED WORK

A. Sports Video Question Answering

Li et al. introduced Sports-QA, a large-scale VideoQA benchmark designed specifically for professional sports [1]. Unlike a number of general VideoQA datasets, Sports-QA includes questions that require a fine-grained understanding of sports actions and features various kinds of temporal reasoning such as description, chronology, causality, and counterfactual reasoning. The authors also put forward an Auto-Focus Transformer that adjusts its attention according to different temporal scales. Sports-QA is directly related to CourtVision, since the basketball videos and questions employed in this project come from this benchmark. However, instead of training the Sports-QA model, CourtVision uses the dataset to assess the impact of various frame-selection strategies on a frozen multimodal vision-language model.

Xiao et al. introduced NExT-QA to study video understanding beyond basic scene description [2]. This benchmark places a special focus on causal and temporal action reasoning. Their experiments showed that models which could answer descriptive questions still had difficulty with questions that required causal or temporal reasoning. This finding is especially pertinent to CourtVision, since the basketball portion used in this study includes both descriptive and causal questions. The distinction between these two categories allows one to examine whether frame-selection methods retain adequate temporal context.

B. Adaptive Frame Selection for Video-LLMs

Zhang et al. proposed Q-Frame, a query-aware frame-selection technique for Video-LLMs [3]. Rather than choosing video frames at random, Q-Frame identifies frames that are relevant to the user’s query by means of image-text matching. It is possible for the method to use a network like CLIP in order to estimate the relationship between individual frames and the text of the question. CourtVision’s third sampling strategy is similar in this sense, because CLIP similarity is used as one element of the frame-selection score. The main difference lies in the fact that CourtVision combines query relevance with motion, player detection, and basketball detection, rather than depending solely on image-text relevance.

Hu et al. proposed an adaptive frame-selection method in which a lightweight multimodal model identifies query-relevant video frames before they are processed by a frozen downstream Video-LLM [4]. While their study points out the drawback of uniform sampling—that important information can occur between the frames that are selected uniformly—CourtVision looks at the same general issue but deals with it by using conventional computer vision signals such as motion and object detection together with CLIP, rather than by training a separate frame-selection model.

C. Vision-Language Similarity and Object Detection

CLIP learns joint visual and text representations from image-text pairs and allows for direct similarity comparisons between image and text embeddings [5]. In CourtVision, CLIP

is used to assess the semantic relevance between a candidate basketball frame and the corresponding Sports-QA question.

The YOLO approach treats object detection as a single problem for prediction by means of a neural network and has thus become a popular family of real-time object detectors [6]. CourtVision makes use of a recent implementation of YOLO11n in order to detect people and sports balls in the candidate frames; the results of this detection are then used as signals to prioritize those frames which contain basketball activity. The initial version of the YOLO method was presented by Redmon et al. at CVPR 2016.

III. METHODS

A. Dataset Preparation

The experiments made use of examples from the Sports-QA dataset [1], which includes professional sports videos together with questions that require various kinds of visual and temporal reasoning.

For this project, 60 basketball question-answer pairs were chosen from the training split. The same 60 questions were used with each frame-selection strategy so that the comparisons could be matched and the only intended experimental variable was the frame-selection method.

The subset had 49 descriptive questions and 11 causal questions. Each example had a video identifier, video path, question, correct answer, and question type.

The results reported should be understood as showing a comparison between different sampling strategies rather than as a measure of performance on a benchmark test set, since the experiments make use of a subset of the Sports-QA training split rather than a held-out test split.

B. Uniform Temporal Sampling

Uniform frame sampling was used as the baseline. For a video containing N frames, $K = 8$ frames were selected at approximately equal temporal intervals:

$$f_i = \text{round} \left(\frac{i(N-1)}{K-1} \right),$$

where $i \in \{0, \dots, K-1\}$.

This approach guarantees broad coverage from the beginning to the end of the video but does not consider motion, objects, or the question being asked.

C. Motion and YOLO-Guided Sampling

The second method tries to choose frames from basketball footage which are visually informative. Instead of looking at each and every frame, it samples the candidate frames at a stride of five. Three features are computed for each candidate.

The motion is first estimated by calculating the mean absolute pixel difference between consecutive candidate frames; before this calculation the frames are converted to grayscale and resized. The larger the difference, the greater the visual change, which could be due to player movement, ball movement, camera motion, or scene transitions.

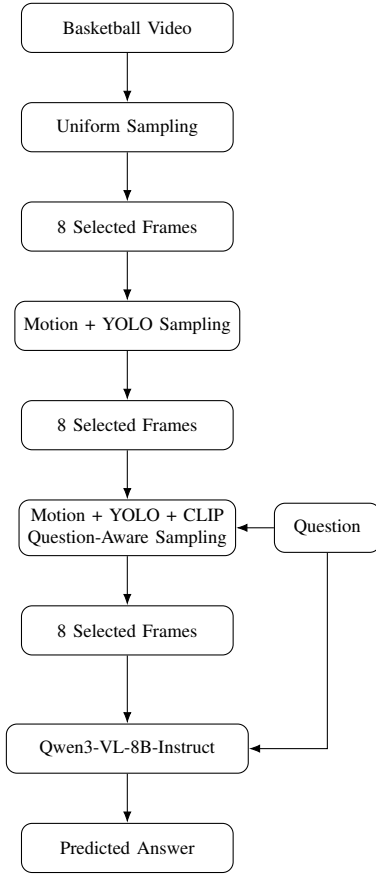


Fig. 1: Overview of the CourtVision VideoQA pipeline. Three frame-selection strategies select eight frames from the same basketball video. Uniform and Motion+YOLO sampling depend only on the video, while Motion+YOLO+CLIP additionally uses the question for query-aware frame selection. The eight selected frames and question are then provided to the same Qwen3-VL-8B-Instruct model to generate the predicted answer.

Second, YOLO11n is used to detect players with the COCO person class, taking into account both the detection confidence and the amount of the image occupied by the detected people.

Third, YOLO is applied to identify the COCO sports ball category, and extra points are awarded when basketball detections are present.

The three features are normalized within each video and combined as

$$S_{MY} = 0.57M + 0.29P + 0.14B,$$

where M , P , and B represent normalized motion, person, and ball scores, respectively.

The YOLO inference process employs a confidence threshold of 0.20 and an image size of 640 pixels. Once the candidate frames have been ranked, a temporal-diversity constraint is applied which stops highly ranked frames that are very close to each other from having a dominant influence on the final

choice. A minimum gap of 15 frames is used, and eight frames are eventually selected.

The method does not depend on the question, so different questions concerning the same video yield the same Motion+YOLO frame set.

D. Motion, YOLO, and CLIP-Guided Sampling

The third method incorporates question-aware semantic relevance into the earlier approach.

Candidate frames are embedded using the pretrained openai/clip-vit-base-patch32 model [5]. The Sports-QA question is separately encoded using CLIP’s text encoder. Both image and text embeddings are normalized, and their dot product is used as cosine similarity.

The CLIP similarity obtained for each candidate frame is normalized together with the other computer vision signals, and the final score is

$$S_{MYC} = 0.40M + 0.20P + 0.10B + 0.30C,$$

where C is normalized CLIP frame-question similarity.

Unlike Motion+YOLO sampling, this method is question dependent. Two QA pairs associated with the same basketball video may therefore produce different sets of eight frames.

The motivation is that motion and detection scores identify visually active basketball moments, while CLIP may prioritize moments whose visual content is semantically related to the specific question.

E. Multimodal Question Answering

The same downstream model, Qwen3-VL-8B-Instruct, is used by all the methods, and no fine-tuning is carried out.

Each example provides the model with exactly eight chosen frames, the frames being resized to 448×252 pixels and the same prompt being used for all the methods:

The images listed are frames taken from a basketball video and are arranged in chronological order. You are to answer the question based solely on the information that is visible in the frames. Just give a brief answer without any explanation.

Generation is deterministic when sampling is disabled, and the number of newly generated tokens is set at 16.

If the number of frames, the image resolution, the model, the prompt, and the decoding configuration are all kept the same, then frame selection becomes the primary experimental variable.

F. Evaluation Metrics

Two main metrics are used.

Exact Match (EM) compares the normalized predictions with the normalized actual answers; to do this, the text is changed to lowercase, punctuation and English articles are deleted, and the white space is made consistent. A prediction is given a score of one only in the case where the normalized strings match exactly.

Token F1 awards partial credit in the case where the predicted answers and the correct answers have tokens in

common. Precision and recall are calculated based on the overlapping answer tokens:

$$P = \frac{|T_p \cap T_g|}{|T_p|}, \quad R = \frac{|T_p \cap T_g|}{|T_g|},$$

and

$$F_1 = \frac{2PR}{P+R}.$$

The same questions are used to evaluate each method, and therefore the pairwise results are assessed by means of an exact McNemar test as well; this allows the outcomes to be determined for cases in which both methods reach the correct answer, cases in which only one of the methods does so, and cases in which neither method does.

G. Basketball Analytics Extension

An analytics component was developed as an extension to the VideoQA experiment, involving the combination of YOLO detections with multi-object tracking in order to assign persistent IDs to the visible players from one frame to the next.

The pixel-space movement statistics and the frame-level spacing between the visible players are calculated using tracked player centres. Since calibration of the broadcast cameras is not carried out, the values are understood in terms of image pixels rather than physical metres.

The player trajectories and player occupancy over the image are also shown in the analytics interface. This illustrates how the same computer-vision pipeline which is used for frame analysis can be employed in basketball-specific visual analytics. However, these features are not part of the primary VideoQA accuracy comparison.

IV. EXPERIMENTS AND RESULTS

A. Overall VideoQA Performance

All three sampling methods were evaluated on the same 60 Sports-QA basketball questions.

TABLE I: Overall VideoQA Performance

Method	EM	Mean F1	Median F1
Uniform	25.00%	0.3961	0.2500
Motion + YOLO	25.00%	0.3929	0.2679
Motion + YOLO + CLIP	21.67%	0.3641	0.2222

Both uniform sampling and Motion+YOLO got 15 out of the 60 questions correct when assessed by Exact Match, so Motion+YOLO did not achieve any overall improvement in exact-match accuracy. Moreover, the mean Token F1 of Motion+YOLO was also very similar to that of uniform sampling, dropping by about **0.0033**.

Adding CLIP question relevance did not improve performance on this evaluation subset: the exact-match accuracy fell from **25.00%** with Uniform and Motion+YOLO to **21.67%**, and the mean Token F1 dropped to 0.3641.

The small difference between Uniform and Motion+YOLO is important, since the two methods did not simply obtain the

same set of correct answers. There were twelve questions that were answered correctly by both approaches, three that were solved only by Motion+YOLO, and three that were solved only by Uniform. The McNemar test carried out exactly gave a result of $p = 1.0$, thereby offering no indication of a systematic difference between the two methods.

When comparing Motion+YOLO+CLIP with Uniform, both methods got 12 questions right, one question was solved only by the CLIP-guided method, and three were solved only by Uniform. The exact McNemar test yielded a p -value of **0.625**. Hence, even though the CLIP-guided sampling performed less well numerically on this subset, the experiment does not show a statistically significant difference.

TABLE II: Paired Exact-Match Results. MY denotes Motion+YOLO, and MYC denotes Motion+YOLO+CLIP.

Comparison	A Only	B Only	p
MY vs. Uniform	3	3	1.000
MYC vs. Uniform	1	3	0.625
MYC vs. MY	1	3	0.625

B. Performance by Question Type

Looking at the results according to question type provides further information regarding the effect of temporal context.

TABLE III: Token F1 by Question Type. MY denotes Motion+YOLO, and MYC denotes Motion+YOLO+CLIP.

Method	Descriptive (49)	Causal (11)
Uniform	0.4491	0.1602
MY	0.4582	0.1017
MYC	0.4113	0.1537

Motion+YOLO achieved a slight improvement in Token F1 for descriptive questions when compared to Uniform sampling, but its Token F1 on causal questions decreased considerably.

The fact that visually significant moments are more useful when a question involves what is visible or what action took place means that picking out individual high-activity frames can result in the loss of the contextual evidence needed to explain why an action occurred.

On the 11 causal questions, all three methods achieved zero exact-match accuracy, which demonstrates that this subset continued to pose a difficulty for the frozen multimodal model no matter what sampling strategy was used.

C. Qualitative Comparison

Looking at some of the frames in detail allows the quantitative results to be understood.

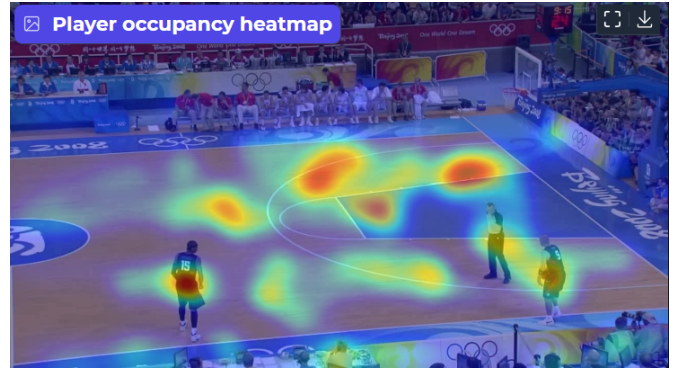
Fig. 3 compares the frames selected by the three sampling methods for a representative basketball question.

Uniform sampling always includes the whole time span of each video; even if some of the selected frames are not visually noteworthy, the model still gets observations from the start, the middle, and the end of the clip.

Motion+YOLO tends to focus on frames that have strong motion or on those in which the players are clearly visible.



(a) Player tracking with persistent track IDs.



(b) Player occupancy heatmap.

Fig. 2: Basketball analytics generated by CourtVision. (a) YOLO and multi-object tracking are used to identify and track players across video frames. (b) Tracked player positions are aggregated to visualize player occupancy in image-space coordinates.

Yet high motion can also result from the camera panning, zooming, or making cuts. Moreover, basketball videos usually have a number of people detected in almost every frame that is considered as a candidate, which decreases the discriminative power of the player feature.

Basketball detection was less reliable, since the ball took up a very small part of most of the broadcast frames; as a result, the frames which contained the most useful basketball action did not always get high ball scores.

In some cases CLIP enhanced the semantic relationship between the question and the individual images selected, but this did not result in better downstream QA performance. There are many questions which require an understanding of the relationship between events rather than the recognition of a visually relevant object or scene in a single frame.

V. DISCUSSION AND SUMMARY

The experiments show that more complex frame-selection methods do not necessarily improve basketball VideoQA performance. Uniform sampling remained a strong baseline because it consistently preserved temporal coverage across the full video. Motion+YOLO achieved the same exact-match accuracy as Uniform and slightly improved descriptive-question Token F1, but it performed worse on causal questions. This suggests that visually active frames may help with recognizing what is happening while still missing the context needed to understand why an event occurred.

The CLIP-based method was intended to make frame selection more relevant to the question, but it produced lower overall performance. One possible explanation is that semantic similarity between a single frame and a question does not guarantee that the frame contains enough evidence to answer the question. For temporal or causal questions, the model may need information from several moments rather than a set of individually relevant frames.

The study has several limitations. Only 60 basketball questions were evaluated, and the subset was drawn from the Sports-QA training split rather than a held-out test set. The

frame-selection weights were manually chosen, only eight frames were provided to the VLM, and the model was not fine-tuned for Sports-QA. In addition, small-object detection of the basketball was sometimes unreliable.

Future work could evaluate the methods on a larger held-out set and explore temporal-window sampling that preserves frames before, during, and after an important event. The basketball analytics module could also be extended with court calibration, team identification, and real-world player-spacing measurements. Overall, the results suggest that preserving temporal context is at least as important as selecting visually or semantically relevant individual frames.

(a) Uniform Sampling



(b) Motion + YOLO



(c) Motion + YOLO + CLIP



Fig. 3: Comparison of selected frames for the question “How does the right team fail to do a 2-point shot?” (a) Uniform sampling, (b) Motion+YOLO sampling, and (c) Motion+YOLO+CLIP question-aware sampling. Frame indices are shown below each selected frame.

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